

THANH TRINH-XUAN

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EDUCATION

Hanoi University of Science and Technology (HUST) <i>Bachelor of Engineering in Electrical Engineering</i> GPA: 3.2/4.0 Research Focus: Microgrid Optimization and Advanced Mathematical Programming in Power Systems. Affiliation: Member of 100 Renewable Energy Laboratory. Relevant Coursework: Power System Analysis, Power System Automation, Power System Protection and Control, Power Plants and Substations, Circuit Theory, Digital System Design, Electronic Design, Programming Techniques.	Hanoi, Vietnam Aug. 2026 (Expected)
Lam Son High School for the Gifted Student <i>High School Diploma, Mathematics Specialization</i>	Thanh Hoa, Vietnam Jul. 2022

RESEARCH EXPERIENCE

Undergraduate Researcher <i>Power Grid and Renewable Energy Laboratory, Hanoi, Vietnam</i> <i>Supervisor: Assoc. Prof. Nguyen Duc Tuyen</i>	Sep. 2024 – Present
Bachelor's Thesis: Resilience-Oriented Peer-to-Peer Energy Trading in Multi-Microgrids via Network-Constrained Distributed Predictive Control - Modeled P-Q-V dynamics using Second-Order Cone Programming (SOCP); mitigated relaxation inexactness (fictitious currents) by integrating load-shedding penalty functions to enforce exactness. - When extreme faults cause network fragmentation and render centralized control unviable, implemented Analytical Target Cascading (ATC) to maintain decentralized coordination. - During blackouts that invalidate static day-ahead dispatch, mitigated immediate power imbalances by designing Model Predictive Control (MPC) frameworks for dynamic rescheduling.	2025 – 2026
Project: Distributed Energy Management Frameworks (ICGEA & JST Research) - Developed a 3-layer architecture linking MILP for discrete unit commitment, ADMM for decentralizing tie-lines without exposing private data, and Nucleolus for fair Peer-to-Peer (P2P) profit sharing. - Optimized local microgrid load scheduling via Demand Side Management (DSM). - Highly constrained P2P parameters caused ADMM divergence; resolved this by implementing an adaptive penalty tuning mechanism based on primal-dual residual balancing to ensure rapid convergence.	2024 – 2025
Lab Contribution & Mentorship - Engineered custom Multi-Agent AI workflows to accelerate personal literature synthesis. - Mentored junior researchers in academic reading methodologies and algorithmic modeling, driving lab research productivity.	2024 – Present

ACADEMIC PROJECTS

Machine Learning for EV Load Forecasting

Mar. 2026 – April 2026

- Developed a "Proxy-Lag Cascade Ensemble" architecture using 2 models XGBoost (t+24h and t+1) to eliminate phase lag in time-series forecasting.

Project 2: Industrial Power Supply & Distribution Network Design

Nov. 2025 – Dec. 2025

- Engineered a comprehensive power supply and distribution network for a large-scale machine tool manufacturing plant.
- Calculated wiring schematics, selected optimal cable cross-sections, and determined precise transformer sizing.
- Performed rigorous short-circuit calculations and designed protection schemes to ensure operational reliability.

Electronic & Hardware Prototyping

2023 – 2024

- Designed, simulated (LTspice/Proteus), and physically constructed an AC current measurement circuit, 7-segment digital display, and Class AB audio amplifier.
- Modeled and simulated a DC-DC Buck converter topology (MATLAB/Simulink) for stable step-down voltage regulation.

PUBLICATIONS & PRESENTATIONS

- [1] **T. X. Thanh (1st Author)**, N. T. Anh, T. X. Hung, N. D. Tuyen, T. T. Son, G. Fujita, "A Decentralized MILP-ADMM Framework for P2P Energy Trading in Interconnected Microgrids," *10th International Conference on Green Energy and Applications (ICGEA)*, Mar. 2026. **[Oral Presentation]**
- [2] L. A. Quan, **T. X. Thanh (2nd Author)**, N. T. Anh, P. T. Vinh, T. X. Hung, N. D. Tuyen, T. T. Son, "Microgrid Optimization with MILP-based Demand Side Management," *Journal of Science and Technology (JST)*, vol. 36, Jun. 2026. **[Oral Presentation at Vietnam Student Forum]**

SKILLS

Optimization: MILP, SOCP, Convex Relaxation

Control & Algorithms: ADMM, ATC, MPC, Game Theory

Programming: Python (Pyomo), C, C++

EE Simulation: Simulink, LTspice, PVsyst, Proteus

Tools & AI: Multi-Agent AI, LaTeX, Git

Languages: English (IELTS 6.0, Target 7.0)

HONORS & AWARDS

10th Int. Conf. on Green Energy and Applications (ICGEA)
Best Presentation Award

Mar. 2026

REFERENCES

Assoc. Prof. Nguyen Duc Tuyen

Associate Professor and Head of PGRE Lab, Power Systems Department, HUST, Vietnam

Adjunct Assoc. Prof, Shibaura Institute of Technology, Japan

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